Yifei Dong

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Education

2025 **Duke University**, *Visiting Scholar*, Robot Dexterity Lab, with Prof. Xianyi Cheng

2023 - Now **KTH Royal Institute of Technology**, *PhD Candidate*, Robotics, Computer Science

2019 - 2022 **ETH Zurich**, *M.Sc.*, Mechanical Engineering

2015 - 2019 **Shanghai Jiao Tong University**, *B.Sc.*, Mechanical Engineering
Tsien Hsue-shen Program & Zhiyuan Honors Program of Engineering

Employment

2023 - Now **Robotics, Perception and Learning, KTH**, *Stockholm*
Doctoral Student, with Prof. Florian Pokorny

2022 **Idiap, EPFL**, *Martigny*, *Research Assistant*, with Dr. Sylvain Calinon
Code contribution: Robotics Code From Scratch

2022 **F&P Robotics**, *Zürich*, *Software Engineer Intern*
Motion planning of door opening and human-robot interaction for a mobile service robot.

2018 **NIO Inc.**, *Shanghai*, *Software Engineer Intern*
LiDAR simulation of an autonomous vehicle in various traffic scenes.

Publications

Journals and Conference Proceedings (* indicates equal contribution):

[J1] Yifei Dong, Xianyi Cheng, Florian Pokorny, *Characterizing Manipulation Robustness through Energy Margin and Caging Analysis*, **RA-L**, 2024, *Paper*

[J2] Haofei Lu, Yifei Dong, Zehang Weng, Florian Pokorny, Jens Lundell, Danica Kragic, *Grasping a Handful: Sequential Multi-Object Dexterous Grasp Generation*, **RA-L**, 2025, *Paper* | *Website*

[C1] Kei Ikemura*, Yifei Dong*, Florian Pokorny, *Latent Diffeomorphic Co-Design of End-Effectors for Deformable and Fragile Object Manipulation*, Proceedings of Robotics: Science and Systems (**RSS**), 2026, *Paper*

[C2] Yifei Dong*, Yan Zhang*, Sylvain Calinon, Florian Pokorny, *Robustness-Aware Tool Selection and Manipulation Planning with Learned Energy-Informed Guidance*, **ICRA**, 2026, *Paper*

[C3] Yifei Dong*, Shaohang Han* et al., *CageCoOpt: Enhancing Manipulation Robustness through Caging-Guided Morphology and Policy Co-Optimization*, **IROS**, 2025, *Paper*

[C4] Yifei Dong, Florian Pokorny, *Quasi-static Soft Fixture Analysis of Rigid and Deformable Objects*, **ICRA**, 2024, *Paper*

[C5] David Cáceres Domínguez et al., *The First WARA Robotics Mobile Manipulation Challenge–Lessons Learned*, European Conference on Mobile Robots, 2025, *Paper* | *Video*

[C6] Yue Chen et al., *DUET: Joint Exploration of User Item Profiles in Recommendation System*, ACL findings, 2026, *Paper*

Under Review:

[1] Yifei Dong, Zhanyi Sun, Lujie Yang, Manuel Baum, Shuran Song, Florian Pokorny, Xianyi Cheng, *Robustness of Robotic Manipulation: Foundations and Frontiers*, under review (IJRR), 2026, *Preprint*

[2] Kei Ikemura*, Yifei Dong* et al., *Sim-to-Real Gentle Manipulation of Deformable and Fragile Objects with Stress-Guided Reinforcement Learning*, tech report, 2025, *Preprint*

[3] Yan Zhang, Yiming Li, Yifei Dong, Florian Pokorny, Sylvain Calinon, *Physics-Informed Eikonal Caging for Whole-Arm Manipulation Planning*, under review, 2026, *Preprint*

- [4] David Blanco-Mulero, Yifei Dong, Julia Borras et al., *T-DOM: A Taxonomy for Robotic Manipulation of Deformable Objects*, under review (RAS), 2025, *Preprint* | *Website*
- [5] Christoffer K. Øhrstrøm, Rafael I. C. Muchacho, Yifei Dong et al., *Parabolic Position Encoding: Vision-Centric, Principled, Extrapolatable, General*, under review, 2026, *Preprint*
- [6] Haofei Lu, Hongjia Liu, Yifei Dong et al., *MoDex: A Diffusion Policy for Sequential Multi-Object Dexterous Grasping*, under review (CoRL), 2026

Selected, Peer-reviewed Workshop Paper:

- [1] Kei Ikemura, Yifei Dong et al. *Efficient End-effector Co-Design by Demonstration for Deformable Fragile Object Manipulation*, RSS Workshop on Robot Hardware-Aware Intelligence (Spotlight), 2025, *Paper*
- [2] Weizhe Ni, Yifei Dong, Xianyi Cheng. *A Taxonomy of Everyday Tool Use for Robotic Manipulation*, RSS Workshop on Data-Centric Robotics: What Data Do Robots Really Need?, 2026
- [3] Yuxiao Zhu, Jinzhou Li, Yifei Dong, Haoyu Li, Xinyuan Luo, Xianyi Cheng. *Belief-Guided Interactive Perception for Manipulation in Clutter under Noisy Proprioception*, RSS Workshop on Planning and Control with Imperfect Sensors and Perception, 2026
- [4] Gawtam Chithra Ramesh, David Blanco-Mulero, Yifei Dong et al. *Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided Vision-Language Models*, IROS Workshop on Robot Manipulation of Deformable Objects (ROMADO), 2025

Academic Activities

Talk, Academic Service, Supervision and Teaching:

- 2026 Main organizer of ICRA 2026 Workshop on manipulation robustness, Vienna | *Website*
- 2025 Guest lecture at Duke University's graduate course, Robotic Manipulation, 2025 Spring
- 2024 Invited talk at Prof. Joel Burdick's group, California Institute of Technology
- 2024 - Now Reviewer for IJRR, RA-L, ICRA, IROS, ISRR
- 2024 - 2026 Supervised master/undergraduate students, Kei Ikemura (next: KTH), Hongjia Liu (next: UCSD), Li Chen, Yuxiao Zhu, Gawtam C. Ramesh, Zhenyuan Liang
- 2024, 2025 Teaching assistant in Introduction to Robotics, DD2410, KTH

References

Prof. Florian T. Pokorny, Professor, KTH Royal Institute of Technology, Sweden. Email: fpokorny@kth.se

Prof. Xianyi Cheng, Assistant Professor, Duke University, USA. Email: xianyi.cheng@duke.edu

Prof. Danica Kragic, Professor, KTH Royal Institute of Technology, Sweden. Email: dani@kth.se

Dr. Sylvain Calinon, Senior Research Scientist, Idiap Research Institute; Lecturer, EPFL, Switzerland. Email: sylvain.calinon@epfl.ch

Experiences

2024 **Mobile Manipulation Challenge, ABB**, *Västerås, Sweden*
Team member of KTH: Custom gripper co-design for multi-task manipulation | Slides

2021 **Robotic Systems Lab, ETH**, *Zürich*, *Graduate Researcher*, with Prof. Marco Hutter
Master thesis: Mobile door state estimation and parameter identification | Thesis
Semester thesis: Contact-implicit MPC for mobile manipulation | Video | Thesis

2020 **Computer Vision and Geometry Lab & Microsoft**, *Zürich*
Course project: Object mesh reconstruction using RGB-D cameras. | Thesis | Git | Video

2019 **SAIC Motor**, *Shanghai*
Bachelor thesis: Strategy optimization of autonomous emergency braking

2018 **Institute of Design and Control Engineering, SJTU**, *Shanghai*
3-DOF plum processing system for surface-curving and core-deprivation. | Video

2017 **Institute of Intelligent Vehicle, SJTU**, *Shanghai*
Structural design and locomotion formulation of a snake robot. | Video

Awards & Grants

2024	Travel Grant from the Karl Engvers Foundation
2021, 2022	ETH Scholarship for International Students
2019	Excellence Award, BAIC Automobile Invitational Tournament
2018	Tan Kah Kee Invention Award
2016, 2018	General Motors Scholarship
2017	Huawei Scholarship
2016, 2017	Zhiyuan College Honors Scholarship

Skills

Robot	Franka Panda, Allegro hand, Ufactory Xarm 7, ANYmal legged robot, etc.
Programming	Python, C++, MATLAB, C, C#
Tools	ROS/ROS2, Genesis, Mujoco, Pybullet, Isaac Gym, Git, Docker, Blender, PyTorch, Codex, Simulink, Unity3D, CAD (ANSYS, SolidWorks, CATIA, NX Unigraphics)
Language	English (TOEFL 109/120), Chinese (native), German (Goethe B1)